```python
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#Student ID.2326310

# as taught in the work shop during the previous classes we install numpy

#we import and callit out in the code
import numpy as np

#  According to the research i took oh w3 schools below is the Function to validate input as a number
def validate_input(prompt):
    while True:
        try:
            value = float(input(prompt))
            return value
        except ValueError:
            print("Invalid input. Please enter a valid number.")

# futher more i used the function create   and range to imput the valuse of the matrix a 3x3 matrix
#from user input and print it out in a matrx form
def create_matrix():
    matrix = []
    for i in range(3):
        row = []
        for j in range(3):
            value = validate_input(f"Input the number of the matric for A[{i+1}][{j+1}]: ")
            row.append(value)
        matrix.append(row)
    return matrix
print("Enter values for a 3x3 Matrix:")
matrix_A = create_matrix()
A = np.array(matrix_A)
print("\nMatrix A:")
print(A)

# Futher more icalculating  the determinant of the matrix
determinant = np.linalg.det(A)
print("\nDeterminant of A:", determinant)

#  to check  if matrix A,determinant is = 0
if determinant != 0:
    # Calculate the inverse of the matrix
```

Shells

Python

```
>>> (executing file "PYTHON question 1.py")
Enter values for a 3x3 Matrix:
Input the number of the matric for A[1][1]: 1
Input the number of the matric for A[1][2]: 2
Input the number of the matric for A[1][3]: 3
Input the number of the matric for A[2][1]: 9
Input the number of the matric for A[2][2]: 8
Input the number of the matric for A[2][3]: 7
Input the number of the matric for A[3][1]: 6
Input the number of the matric for A[3][2]: 5
Input the number of the matric for A[3][3]:
Invalid input. Please enter a valid number.
Input the number of the matric for A[3][3]: 4

Matrix A:
[[1. 2. 3.]
 [9. 8. 7.]
 [6. 5. 4.]]

Determinant of A: 0.0
```

Source structure

validate_input()
create_matrix()

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```python
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print("\nDeterminant of A:", determinant)

#  to check  if matrix A,determinant is = 0
if determinant != 0:
    # Calculate the inverse of the matrix
    inverse_A = np.linalg.inv(A)
    print("\nInverse of A:")
    print(inverse_A)
# the program calculates A * A and A * A * A using matrix multiplication and prints the results.
A_squared = np.dot(A, A)
A_cubed = np.dot(A_squared, A)
print("\nA * A:")
print(A_squared)
print("\nA * A * A:")
print(A_cubed)
count = np.sum(A >= 10)
print("\nNumber of values in A >= 10:", count)
```

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PYTHON question3.py | PYTHON question 1.py | PYTHON question 2.py

```python
#Name; Oladipupo Roland Familua
#Student Id. 2326310


import numpy as np
#According to my research the function or operator called trace takes a square matrix as input and uses
the np.trace function from numpy to calculate the trace of the matrix, which is the sum of the elements
on the main diagonal. The result is then returnd it is usally denoted as Calculate_Trace
def calculate_trace(matrix):
    trace = np.trace(matrix)
    return trace

#below is to find the race for 3x3 matrix
matrix_3x3 = np.array([[9, 8, 7], [6, 5, 4], [3, 2, 1]])
trace_3x3 = calculate_trace(matrix_3x3)
print("Trace of 3x3 matrix:", trace_3x3)

# below is to find the race for 4x4 matrix
natrix_4x4 = np.array([[1, 2, 3, 4], [5, 6, 7, 8], [9, 10, 11, 12], [13, 14, 15, 16]])
trace_4x4 = calculate_trace(matrix_4x4)
rint("Trace of 4x4 matrix:", trace_4x4)

below is to find the race for 6x6 matrix
atrix_6x6 = np.array([[1, 2, 3, 4, 5, 6], [7, 8, 9, 10, 11, 12], [13, 14, 15, 16, 17, 18],
                      [19, 20, 21, 22, 23, 24], [25, 26, 27, 28, 29, 30], [31, 32, 33, 34, 35, 36]])
race_6x6 = calculate_trace(matrix_6x6)
rint("Trace of 6x6 matrix:", trace_6x6)
```

Python

```
>>> (executing file "PYTHON question 2.py")
Trace of 3x3 matrix: 15
Trace of 4x4 matrix: 34
Trace of 6x6 matrix: 111

>>>
```

Source structure   File Browser

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PYTHON question3.py | PYTHON question 1.py | PYTHON question 2.py

```python
#Name; Oladipupo Roland Familua
#Student ID; 2326310



def ackermann(m: int, n: int) -> int:
#following are Ackermann's function.
#m as int is the First parameter of Ackermann's function.
#n as int is the Second parameter of Ackermann's function.
    if m == 0:
        return n + 1
    elif m > 0 and n == 0:
        return ackermann(m - 1, 1)
    elif m > 0 and n > 0:
        return ackermann(m - 1, ackermann(m, n - 1))


# Test the function
m = 2
n = 3
result = ackermann(m, n)
print(f"A({m}, {n}) = {result}")
```

Shells

Python

```
>>> (executing file "PYTHON question3.py")
A(2, 3) = 9

>>>
```

Source structure

ackermann()

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